

Otavio Rosa

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EDUCATION

University of South Florida

Jul 2026

B.S., Computer Science

- **GPA:** 3.9/4.0
- **Coursework:** Software Engineering, Algorithms, Computer Vision, Artificial Intelligence, Data Structures

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#, JavaScript, HTML/CSS
- **CV & Data Libraries:** NumPy, OpenCV, Matplotlib, PyTorch, Scikit-learn
- **Frameworks & Tech:** SQL, .NET, Angular, React, Node, Git, Linux, MacOS, Windows, ROS2
- **Software Development Practices:** Object-oriented Programming, Unit Testing, Clean Code
- **Languages:** English (Fluent), Portuguese (Native), Spanish (Upper-intermediate), Japanese (Lower-intermediate)

EXPERIENCE

RARE Lab

Nov 2025 - Present

Undergraduate Researcher

Tampa, FL

- Engineered real-time projector compensation algorithms using OpenCV, PyTorch, and ROS2 for the Unitree Go2 quadruped robot, improving projection accuracy on irregular surfaces to aid navigation in simulated disaster environments.
- Co-authored research manuscripts and conducted literature reviews on advanced projection-mapping techniques with PhD mentors, producing a conference paper that guided the development of the robot's projection system
- Architected a custom Ubuntu 20.04 virtualization workflow on Apple Silicon to resolve networking incompatibilities, reducing development environment setup time and enabling seamless communication with robot peripherals.

ClearSet.AI

May 2025 - Dec 2025

Software Engineer Intern

Tampa, FL

- Developed 30+ responsive and reusable UI components in Angular and TypeScript using object-oriented programming principles, reducing new feature development time by an estimated 20%.
- Authored and maintained comprehensive SQL scripts for 4 major feature rollouts and 10+ data fixes, following clean code standards and ensuring zero-downtime deployments that impacted thousands of user records.
- Identified and resolved 50+ bugs across the full application stack, from the C# .NET backend to the Angular frontend, implementing unit testing to improve system reliability and contributing to a 15% decrease in critical support tickets.

PROJECTS

Please Don't Scroll | [Please Don't Scroll](#)

Nov 2025 - Present

Startup

- Designed and deployed a public-facing wellness platform aimed at reducing dopamine addiction, establishing a distinct brand identity with custom typography and color theory.
- Architecting a cross-platform ecosystem (Web, Chrome Extension, Mobile) to track user engagement metrics; currently beta-testing the React.js web portal.
- Developed a high-performance landing page using semantic HTML5 and vanilla CSS3, prioritizing rapid load times and minimal payload size by eliminating framework overhead.

Object Finder | [Object Finder](#)

Oct 2025

- Engineered an object detection pipeline using Python, OpenCV, and NumPy to identify an object's precise position, scale, and orientation in high-resolution images.
- Dramatically optimized the keypoint matching algorithm by vectorizing distance calculations with NumPy broadcasting, slashing execution time from over 2 minutes to under 2 seconds and meeting strict 20-second performance requirements.

HONORS AND AWARDS

- 8x Dean's List (2022-2025)
- Presidential Honors Scholarship and Honors Medallion
- National MUN 2023 Best Position Paper Award

ORGANIZATIONS

Google Developer Student Club

Community Lead

- Led the planning and execution of HackUSF, a hackathon with 200+ participants, coordinating logistics, timelines, and resources to ensure a seamless event.